



NTNU

TMA4125 Matematikk 4N

#4 – Nonlinear Equations & Newton's method

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Introduction / Recap.

Recap 1. For a given function f we want to consider the equation

$$f(x) = 0 \tag{1}$$

and find its solutions x^* , which we also call **roots** to the equation.

Challenge. In many applications:

Often these (again) might be complicated/impossible to solve analytically

⇒ Develop **numerical techniques** to solve (1).

Scalar and systems of equations

We consider **scalar** equations first, i. e. $f: \mathbb{R} \rightarrow \mathbb{R}$, with just one equation and one variable, for example

$$x^3 + x^2 - 3x - 3 = 0.$$

After that we will consider **systems of equations**, for example

$$\begin{aligned} xe^y &= 1, \\ -x^2 + y &= 1. \end{aligned}$$

We can write this also as a functions $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$, which in this example would be $n = 2$ and

$$f(x, y) = \begin{pmatrix} xe^y - 1 \\ -x^2 + y - 1 \end{pmatrix} \quad \text{and we want to solve} \quad f(x, y) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Recap 2. Fix point iteration.

Idea. reformulate $f(x) = 0$ into a fix point equation $g(x) = x$.

Algorithm. Fix point iteration

Input Given a function g and a starting value $x^{(0)}$

1. For $k = 0, 1, 2, \dots$ compute

$$x^{(k+1)} = g(x^{(k)}).$$

and we discussed its convergence:

Theorem. If there is an interval $[a, b]$ such that $g \in C^1[a, b]$, $g([a, b]) \subset (a, b)$ and there exist a positive constant $L < 1$ such that $|g'(x)| \leq L < 1$ for all $x \in [a, b]$, then

- ▶ g has a unique fixed point x^* in (a, b) .
- ▶ The fix point iterations $x^{(k+1)} = g(x^{(k)})$ converges towards x^* for all starting values $x^{(0)} \in [a, b]$.

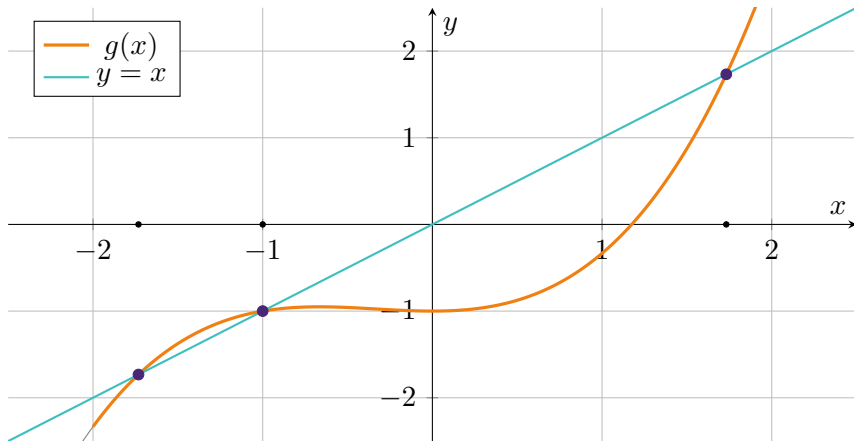
We obtained linear convergence: $|x^* - x^{(k+1)}| \leq |x^* - x^{(k)}|L$ for all k .

Fix point equation: Rearrange f to find g

We rewrite

$$f(x) = x^3 + x^2 - 3x - 3 = 0 \quad \text{to} \quad x = \frac{x^3 + x^2 + 3}{3} = g(x).$$

Interpretation. Fixed points are intersections of $g(x)$ with $y = x$.

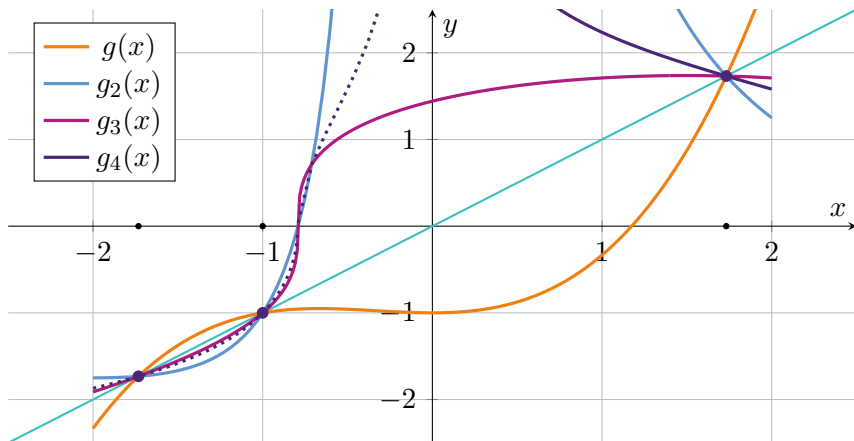


Fix point equation: g is not unique

Other possibilities:

$$g_2(x) = \frac{-x^2 + 3x + 3}{x^2}, \quad g_3(x) = \sqrt[3]{3 + 3x - x^2}, \quad g_4(x) = \sqrt{\frac{3 + 3x - x^2}{x}}$$

Challenge. all of them have different g' and hence different L .



Newton's method – a first idea.

Observation.

Locally around a fix point x^* a small $g'(x^*)$ is preferable (small L).
 $\Rightarrow$ What happens if we could (can) choose g such that $g'(x^*) = 0$?

We use $e^{(k)} = x^* - x^{(k)}$ for the iterates of the fix point algorithm and a Taylor expansion (around x^*)

$$\begin{aligned} e^{(k+1)} &= x^* - x^{(k+1)} = g(x^*) - g(x^{(k)}) = g(x^*) - g(x^* - e^{(k)}) \\ &= -g'(x^*)e^{(k)} + \frac{1}{2}g''(\xi_k)(e^{(k)})^2, \quad (\xi_k \text{ between } x^* \text{ and } x^{(k)}) \end{aligned}$$

Assume we choose $g'(x^*) = 0$ and that we have a constant M such that $|g''(x)|/2 \leq M$. Then

$$|e^{(k+1)}| \leq M|e^{(k)}|^2$$

Also known as **quadratic convergence**.

How to find our favourite fix point equation.

Question. Given an equation

$$f(x) = 0$$

with an unknown solution x^* .

Can we find a g with fixed point $x^* = g(x^*)$ and $g'(x^*) = 0$?

Idea. For *any* $h(x)$ we have due to $f(x^*) = 0$ that

$$g(x) = x - h(x)f(x) = x - \frac{f(x)}{f'(x)}$$

has a fixed point $x^* = g(x^*)$. The derivate reads using the product rule

$$g'(x) = 1 - h'(x)f(x) - h(x)f'(x)$$

At the fixed point we get

$$g'(x^*) = 1 - h(x^*)f'(x^*) - \overset{0}{h'(x^*)f(x^*)}$$

$\Rightarrow$ Choose $h(x) = 1/f'(x)$ we achieve $g'(x^*) = 0$.

Newton's method.

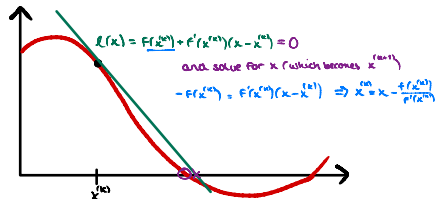
Algorithm.

Input a function f , its derivative f' , and a start point $x^{(0)}$.

1. For $k = 0, 1, 2, \dots$ compute

$$x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f'(x^{(k)})}$$

Geometric interpretation of Newton's met.



Error analysis

We constructed the method to give quadratic convergence

$$|e^{(k+1)}| \leq M |e^{(k)}|^2,$$

where $e^{(k)} = x^* - x^{(k)}$.

But under which conditions can we say something about the size of M ?

Let's compare the Taylor series around x^* with Newton's method

$$0 = f(x^{(k)}) + f'(x^{(k)})(x^* - x^{(k)}) + \frac{1}{2}f''(\xi_k)(x^* - x^{(k)})^2 \quad (\text{Taylor series})$$

$$0 = f(x^{(k)}) + f'(x^{(k)})(x^{(k+1)} - x^{(k)}), \quad (\text{Newton's method})$$

where ξ_k is between x^* and $x^{(k)}$. Subtracting both yields

$$f'(x^{(k)}) \underbrace{(x^* - x^{(k+1)})}_{e^{(k+1)} = O(x^{k+1})} + \frac{1}{2}f''(\xi_k) \underbrace{(x^* - x^{(k)})^2}_{e^{(k)} = O(x^k)} = 0 \quad \Rightarrow \quad e^{(k+1)} = -\frac{1}{2} \frac{f''(\xi_k)}{f'(x^{(k)})} (e^{(k)})^2$$

$\Rightarrow$ we need $f'(x^{(k)}) \neq 0$, $f \in C^2[a, b]$ and $x^{(0)}$ sufficiently close to x^* .

Convergence of Newton's method

Theorem. Assume that the function f has a root x^* , and let $I_\delta = [x^* - \delta, x^* + \delta]$ for some $\delta > 0$.

Assume further that

- ▶ $f \in C^2(I_\delta)$.
- ▶ There is a $M > 0$ such that $\left| \frac{f''(y)}{f'(x)} \right| \leq 2M$, for all $x, y \in I_\delta$.

In this case, Newton's method converges quadratically,

$$|e^{(k+1)}| \leq M|e^{(k)}|^2$$

for all starting values satisfying $|x^{(0)} - x^*| \leq \min\{1/M, \delta\}$.

Systems of equations

Instead of **one** function f with **one** variable x we now consider

Systems of nonlinear equations.

$$f_1(x_1, x_2, \dots, x_n) = 0$$

$$f_2(x_1, x_2, \dots, x_n) = 0$$

$$\vdots$$

$$f_n(x_1, x_2, \dots, x_n) = 0$$

We can write this in short as

$$\mathbf{f}(\mathbf{x}) = \mathbf{0},$$

where $\mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^n$.

Example.

Example. Consider the two equations

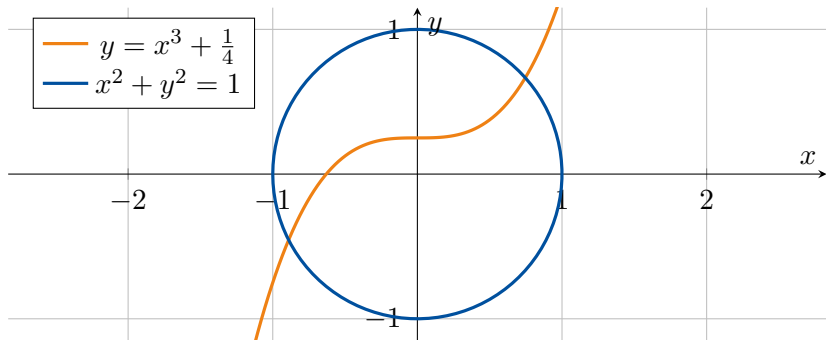
$$x^3 - y + \frac{1}{4} = 0$$

$$x^2 + y^2 - 1 = 0$$

Interpretation.

Rewrite first equation to $y = x^3 + \frac{1}{4} \Rightarrow$ solutions lie on this graph.

Second equation means $1 = x^2 + y^2 \Rightarrow$ point with distance 1 from origin



Towards Newton's method for systems of equations I

Idea. Extend fixed point iterations to systems of equations.

We concentrate on **Newton's method** and the case $n = 2$:

$$\begin{aligned} f(x, y) &= 0 \\ g(x, y) &= 0 \end{aligned} \quad \begin{array}{l} \text{We linearize } f \text{ and } g \\ \Rightarrow \text{tangent plane} \end{array}$$

to avoid getting lost in indices.

Notation. We denote a solution (root) to these by $\mathbf{r} = (x^*, y^*)^T$.

Let $\hat{\mathbf{x}} = (\hat{x}, \hat{y})^T$ that approximates $\mathbf{r}$.

$\Rightarrow$ Search for a better approximation by **linearizing $\mathbf{f}(\mathbf{x}) = \mathbf{0}$** .

In other words: use the multivariate Taylor expansion around $\hat{\mathbf{x}}$

$$f(x, y) = f(\hat{x}, \hat{y}) + \frac{\partial f}{\partial x}(\hat{x}, \hat{y})(x - \hat{x}) + \frac{\partial f}{\partial y}(\hat{x}, \hat{y})(y - \hat{y}) + \dots$$

$$g(x, y) = g(\hat{x}, \hat{y}) + \frac{\partial g}{\partial x}(\hat{x}, \hat{y})(x - \hat{x}) + \frac{\partial g}{\partial y}(\hat{x}, \hat{y})(y - \hat{y}) + \dots$$

where we omit higher order terms, which are small for $(x, y) = \mathbf{x} \approx \hat{\mathbf{x}}$.

Towards Newton's method for systems of equations II

Idea. Ignore the higher order terms and solve

$$f(\hat{x}, \hat{y}) + \frac{\partial f}{\partial x}(\hat{x}, \hat{y})(x - \hat{x}) + \frac{\partial f}{\partial y}(\hat{x}, \hat{y})(y - \hat{y}) = 0$$

$$g(\hat{x}, \hat{y}) + \frac{\partial g}{\partial x}(\hat{x}, \hat{y})(x - \hat{x}) + \frac{\partial g}{\partial y}(\hat{x}, \hat{y})(y - \hat{y}) = 0$$

for x and y as a better approximation (or precise: next iterate).

More compact we can write

$$\mathbf{f}(\hat{\mathbf{x}}) + J(\hat{\mathbf{x}})(\mathbf{x} - \hat{\mathbf{x}}) = \mathbf{0},$$

where $J(\mathbf{x})$ denotes the **Jacobian of $\mathbf{f}$** given by

$$J(\mathbf{x}) = \begin{pmatrix} \frac{\partial f}{\partial x}(x, y) & \frac{\partial f}{\partial y}(x, y) \\ \frac{\partial g}{\partial x}(x, y) & \frac{\partial g}{\partial y}(x, y) \end{pmatrix}$$

Newton's method for systems of equations

Algorithm.

Input A function $f(\mathbf{x})$, its Jacobian $J(\mathbf{x})$ and a starting value $\mathbf{x}^{(0)}$.

1. For $k = 0, 1, 2, \dots$

1.1 Solve the linear system $J(\mathbf{x}^{(k)})\Delta^{(k)} = -f(\mathbf{x}^{(k)})$

1.2 Set $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \Delta^{(k)}$

Note. This can be generalized to n equations with n unknowns, where the Jacobian reads

$$J(\mathbf{x}) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(\mathbf{x}) & \frac{\partial f_1}{\partial x_2}(\mathbf{x}) & \cdots & \frac{\partial f_1}{\partial x_n}(\mathbf{x}) \\ \frac{\partial f_2}{\partial x_1}(\mathbf{x}) & \frac{\partial f_2}{\partial x_2}(\mathbf{x}) & \cdots & \frac{\partial f_2}{\partial x_n}(\mathbf{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1}(\mathbf{x}) & \frac{\partial f_n}{\partial x_2}(\mathbf{x}) & \cdots & \frac{\partial f_n}{\partial x_n}(\mathbf{x}) \end{pmatrix}$$

Example.

To solve our example from before

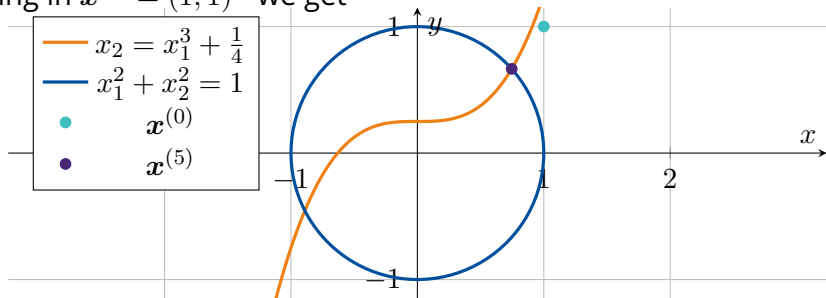
$$f(x, y) = x^3 - y + \frac{1}{4} = 0$$

$$g(x, y) = x^2 + y^2 - 1 = 0$$

we need its Jacobian

$$J(x, y) = \begin{pmatrix} 3x^2 & -1 \\ 2x & 2y \end{pmatrix}$$

Starting in $x^{(0)} = (1, 1)^T$ we get



Error analysis for multivariate Newton's method

Error and convergence analysis for the multivariate case is beyond the scope of this lecture.

Summary. If f is sufficiently differentiable, and there is a solution x^* of the system $f(x) = 0$ with $J(x^*)$ nonsingular, then the Newton iterations will converge quadratically towards x^* for all $x^{(0)}$ sufficiently close to x^* .

Exercise. Solve the other problem from the into

$$f(x, y) = \begin{pmatrix} xe^y - 1 \\ -x^2 + y - 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$